

Decent Coaching Classes
X SB
Academic Conquest - IV
ANSWER KEY & MARKING SCHEME

Subject: Science	Time: 1 Hour	Marks: 30
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Q.1 (A) Correct alternatives (1 mark each) **[5]**

- (b) latent heat of fusion.
- (b) 4 °C — water has minimum volume and hence maximum density at 4 °C.
- (d) 100% — at the dew point the air is saturated with vapour.
- (b) bends away from the normal.
- (c) red — it has the longest wavelength and bends the least; violet bends the most.

Q.1 (B) One-sentence answers, any 3 (1 mark each) **[3]**

- The specific heat of a substance is the amount of heat energy required to raise the temperature of a unit mass of the substance by 1 °C.
- In an isolated system, the heat energy lost by the hot object is equal to the heat energy gained by the cold object.
- For a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant ($\sin i \div \sin r = n$).
- The separation of white light into its component colours while passing through a medium (such as a prism) is called dispersion of light.

Q.2 Scientific reasons, any 2 (2 marks each) **[4]**

- When the temperature of water falls from 4 °C to 0 °C, it expands, so its density decreases; hence the colder water stays at the top and freezes to form an ice layer [1]. The water below the ice layer remains at about 4 °C (liquid), so aquatic life survives [1].
- The surface of the bottle is cold, so the air touching it is cooled below its dew point and becomes saturated [1]. The excess water vapour condenses into tiny water droplets on the bottle, in the same way as dew forms on leaves in the morning [1].
- Stars are point sources of light. The refractive index of the atmosphere keeps changing, so the apparent position and brightness of a star keep changing, making it twinkle [1]. Planets are much nearer and act as a collection of many point sources; the variations average out, so their position and brightness remain almost steady and they do not twinkle [1].

Q.3 Answer in brief, any 3 (2 marks each) **[6]**

- Regelation (definition 1 + activity 1)

Definition: The phenomenon in which ice melts under applied pressure (its melting point falls below 0 °C) and re-freezes when the pressure is removed is called regelation.

Activity: Place a slab of ice on a stand, put a thin metal wire over it and hang equal weights from both ends of the wire. The wire gradually cuts through the slab and comes out from the bottom, yet the slab remains in one piece, because the water formed under the wire re-freezes above it.

- Humidity (definitions 1 + formula 1)

Absolute humidity: the mass of water vapour present in a unit volume of air (measured in kg/m³).

Relative humidity: the ratio of the actual mass of water vapour in a given volume of air to the mass of vapour needed to saturate the same volume of air at that temperature.

Formula: % relative humidity = (actual mass of vapour in a given volume of air ÷ mass of vapour needed to saturate that volume at the same temperature) × 100.

- Mirage (meaning 1 + formation 1)

Meaning: An optical illusion in which water seems to be present on a hot road or in a desert; light from a distant object appears to come from its image inside the ground.

Formation: Air near the hot surface is hotter and rarer than the air above, and the refractive index of air increases with height. Light rays from a distant object are refracted continuously as they pass through these layers and reach the observer as if coming from an image below the ground.

4. Total internal reflection (2×1 mark)

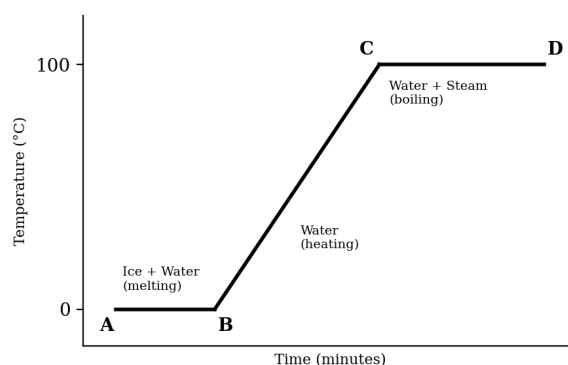
Total internal reflection: When light passes from a denser to a rarer medium and the angle of incidence is greater than the critical angle, the whole of the light is reflected back into the denser medium.

Critical angle: The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° .

Q.4 Answer in detail, any 2 (3 marks each)

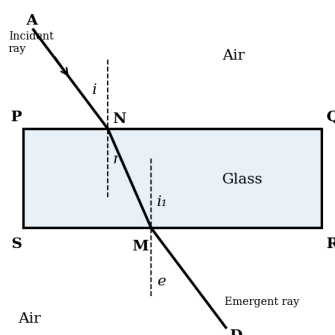
[6]

1. Temperature–time graph (graph 1 + explanation 2)



1. AB (ice $\rightarrow$ water): ice absorbs heat and melts at the constant temperature 0°C , the melting point. The heat absorbed at constant temperature is the latent heat of fusion (80 cal/g for ice).
2. BC: once all the ice has melted, the temperature of the water rises from 0°C to 100°C .
3. CD (water $\rightarrow$ steam): at 100°C , the boiling point, the water absorbs heat but its temperature stays constant until all of it becomes steam. This heat is the latent heat of vaporization (540 cal/g for water).

2. Glass slab: $i = e$ (ray diagram 1 + proof 2)



1. Let AN be the incident ray at N on face PQ, NM the refracted ray inside the glass and MD the emergent ray at face SR. At N (air $\rightarrow$ glass), if n is the refractive index of glass with respect to air: $\sin i \div \sin r = n$. [1]
2. Faces PQ and SR are parallel, so the normals at N and M are parallel and the angle of incidence at M is $i_1 = r$ (alternate angles). [$\frac{1}{2}$]
3. At M (glass $\rightarrow$ air) the refractive index of air with respect to glass is $1/n$. So $\sin i_1 \div \sin e = 1/n$, i.e. $\sin r \div \sin e = 1/n$, giving $\sin e = n \sin r = \sin i$. [1]
4. Therefore $e = i$. Hence the emergent ray MD is parallel to the incident ray AN, but it is laterally displaced. [$\frac{1}{2}$]

3. Rainbow (explanation 2 + diagram 1)

- A rainbow is seen mainly after rainfall. The tiny water droplets in the air act as small prisms.

- Sunlight entering a droplet is refracted and dispersed into its colours. The dispersed light then undergoes total internal reflection at the back surface of the droplet.
- On coming out of the droplet the light is refracted once again. The different colours leave at different angles, so the observer sees a spectrum of seven colours (VIBGYOR).
- Thus a rainbow is the combined effect of refraction, dispersion and total internal reflection.
- Diagram: a sunlight ray entering a spherical water droplet (refraction + dispersion), internal reflection at the back, and coloured rays leaving the droplet (refraction) — with the colours labelled.

Q.5 Numericals, any 2 (3 marks each)

[6]

1. Heat to melt ice: $Q_1 = m \times L = 200 \times 80 = 16\,000 \text{ cal}$ [1]. Heat to warm water from 0°C to 40°C : $Q_2 = m \times c \times \Delta T = 200 \times 1 \times 40 = 8\,000 \text{ cal}$ [1]. Total heat required = $Q_1 + Q_2 = 24\,000 \text{ cal}$ (= 24 kcal) [1].
2. Let the final temperature be T . Heat lost by copper = $200 \times 0.1 \times (100 - T) = 20(100 - T)$ [1]. Heat gained by water = $300 \times 1 \times (T - 20)$ [1]. By the principle of heat exchange: $2000 - 20T = 300T - 6000 \Rightarrow 320T = 8000 \Rightarrow T = 25^\circ\text{C}$ [1].
3. (i) $n = c \div v \Rightarrow v = c \div n = (3 \times 10^8) \div (3/2) = 2 \times 10^8 \text{ m/s}$ [$1\frac{1}{2}$]. (ii) Refractive index of glass with respect to water = $n(\text{glass}) \div n(\text{water}) = (3/2) \div (4/3) = 9/8 \approx 1.125$ [$1\frac{1}{2}$].